

CS543 · Web Services · Assignment 4

Part B — Publish the Contract

Selected Service: Orders

This document provides the OpenAPI 3.0 contract for the CampusEats Orders service. It follows the Assignment 4 requirements: paths, reusable schemas, request parameters/bodies, documented success and failure responses, and an idempotency header for the create operation.

B1–B3: openapi.yaml

```
openapi: 3.0.3
info:
  title: CampusEats Orders Service
  version: 1.0.0
  description: REST API for creating and managing CampusEats orders.

servers:
  - url: http://localhost:5000

paths:
  /orders:
    post:
      summary: Create an order
      operationId: createOrder
      parameters:
        - $ref: '#/components/parameters/IdempotencyKey'
      requestBody:
        required: true
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/CreateOrderRequest'
      responses:
        '201':
          description: Order created
          headers:
            Location:
              description: URI of the created order
              schema:
                type: string
          content:
            application/json:
              schema:
                $ref: '#/components/schemas/OrderResponse'
        '400':
          $ref: '#/components/responses/BadRequest'
        '409':
          $ref: '#/components/responses/Conflict'
        '422':
          $ref: '#/components/responses/UnprocessableEntity'

    get:
      summary: List orders
      operationId: listOrders
      parameters:
        - name: status
          in: query
          required: false
          schema:
            $ref: '#/components/schemas/OrderStatus'
      responses:
        '200':
          description: Matching orders
          content:
            application/json:
              schema:
                type: array
                items:
                  $ref: '#/components/schemas/OrderResponse'
        '400':
          $ref: '#/components/responses/BadRequest'

  /orders/{orderId}:
    get:
      summary: Get one order
      operationId: getOrder
      parameters:
        - $ref: '#/components/parameters/OrderId'
      responses:
```

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    '200':
      description: Order found
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/OrderResponse'
    '404':
      $ref: '#/components/responses/NotFound'

/orders/{orderId}/cancellation:
  post:
    summary: Cancel an order
    operationId: cancelOrder
    parameters:
      - $ref: '#/components/parameters/OrderId'
    requestBody:
      required: false
      content:
        application/json:
          schema:
            $ref: '#/components/schemas/CancellationRequest'
    responses:
      '200':
        description: Order cancelled
        content:
          application/json:
            schema:
              $ref: '#/components/schemas/OrderResponse'
      '400':
        $ref: '#/components/responses/BadRequest'
      '404':
        $ref: '#/components/responses/NotFound'
      '409':
        $ref: '#/components/responses/Conflict'
      '422':
        $ref: '#/components/responses/UnprocessableEntity'

components:
  parameters:
    OrderId:
      name: orderId
      in: path
      required: true
      schema:
        type: string
        example: ORD-20260908-001

    IdempotencyKey:
      name: Idempotency-Key
      in: header
      required: true
      description: Unique key used to safely retry order creation.
      schema:
        type: string
        minLength: 1
        example: 9f6c2b1e-create-order-001

  schemas:
    CreateOrderRequest:
      type: object
      required:
        - customerId
        - items
      properties:
        customerId:
          type: string
          example: STU-1024
        items:
          type: array
          minItems: 1
          items:
            $ref: '#/components/schemas/OrderItemInput'

    OrderItemInput:
      type: object
      required:
        - itemId
        - quantity
      properties:
        itemId:
          type: string
          example: ITEM-101
        quantity:
          type: integer
          minimum: 1
          example: 2

    CancellationRequest:
      type: object

```

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    properties:
      reason:
        type: string
        example: Customer requested cancellation

OrderResponse:
  type: object
  required:
    - orderId
    - customerId
    - items
    - status
    - totalAmount
  properties:
    orderId:
      type: string
      example: ORD-20260908-001
    customerId:
      type: string
      example: STU-1024
    items:
      type: array
      items:
        $ref: '#/components/schemas/OrderItem'
    status:
      $ref: '#/components/schemas/OrderStatus'
    totalAmount:
      type: number
      format: double
      example: 438.50
    createdAt:
      type: string
      format: date-time

OrderItem:
  type: object
  required:
    - itemId
    - quantity
    - unitPrice
  properties:
    itemId:
      type: string
      example: ITEM-101
    quantity:
      type: integer
      minimum: 1
      example: 2
    unitPrice:
      type: number
      format: double
      example: 219.25

OrderStatus:
  type: string
  enum:
    - PENDING
    - CONFIRMED
    - CANCELLED
    - COMPLETED

Problem:
  type: object
  required:
    - type
    - title
    - status
    - detail
  properties:
    type:
      type: string
      example: https://campuseats.example.com/problems/invalid-order
    title:
      type: string
      example: Invalid order
    status:
      type: integer
      example: 400
    detail:
      type: string
      example: The request body is malformed.

responses:
  BadRequest:
    description: Malformed request
    content:
      application/json:
        schema:
          $ref: '#/components/schemas/Problem'

```

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NotFound:
  description: Order does not exist
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Problem'

Conflict:
  description: Order state conflicts with the requested operation
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Problem'

UnprocessableEntity:
  description: Request is valid but refused by the Orders domain
  content:
    application/json:
      schema:
        $ref: '#/components/schemas/Problem'

```

Endpoint Summary

POST /orders — Create a new order

Success: 201 Failure: 400, 409, 422

GET /orders/{orderId} — Read one order

Success: 200 Failure: 404

GET /orders?status=... — Filter/list orders

Success: 200 Failure: 400

POST /orders/{orderId}/cancellation — Change order state to cancelled

Success: 200 Failure: 400, 404, 409, 422

Why this contract satisfies Part B

- The contract is written before handler code and contains info, servers, paths and components.schemas.
- Reusable request/response shapes are declared once under components.schemas and referenced with \$ref.
- Every operation documents its success response and its applicable failure responses.
- The create operation declares the required Idempotency-Key header so a retry can return the original result instead of creating a duplicate order.
- All failures use the same Problem shape with type, title, status and detail.

Validation

Assignment 4 requires this openapi.yaml to be validated with editor.swagger.io or openapi-spec-validator and the zero-error result to be included in the final submission. That validation output is not included in this PDF because it must be generated from the actual saved YAML file.